Automated Student Outreach & AR Intelligence Workflow

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Tools: Excel Office Scripts (TypeScript), Excel VBA
Domain: Operations Analytics, Workflow Automation, Data Engineering for Outreach

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Executive Overview

This project is a fully automated, production-ready workflow designed to transform raw operational data into **actionable, auditable email outreach lists**, while simultaneously **updating source systems to reflect completed contact activity**.

The automation replaces a previously manual, error-prone process spanning:

- data cleanup,
- compliance filtering,
- prioritization logic,
- list building for different outreach types,
- and post-outreach system updates.

The result is a **single-run pipeline** that:

1. Prepares and standardizes dashboard data
2. Derives AR-specific and non-AR outreach segments
3. Generates email-ready lists with correct messaging
4. Enforces volume caps and prioritization rules
5. Writes contact dates back to the source dashboard for tracking

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High-Level Workflow Architecture

1. **Report Normalization & Enrichment**
    - Clean incoming dashboard exports
    - Remove irrelevant rows and columns
    - Enrich records with lookup-based metadata

- Create decision helper columns

2. **AR Filtering & Staging**
    - Persist a fully processed "AR filters" worksheet
    - Add logic-driven flags (e.g., AR ≥ 100)
3. **Personal Outreach List (Michael Larrauri)**
    - Build two distinct segments:
        - AR ≥ 100
        - Non-AR but eligible
    - Apply color-coded visual cues
    - Attach correct email templates
4. **Team Outreach List**
    - Prioritize new records
    - Backfill older records as needed
    - Enforce a hard daily cap
    - Populate templates automatically
5. **Feedback Loop**
    - Update contact dates in the source dashboard
    - Ensure records reflect outreach completion

Script-by-Script Breakdown

Script 1 — Report Automation & AR Filter Engine

(Primary data engineering and enrichment script)

Purpose

This script converts a raw dashboard export into a **clean, enriched, decision-ready dataset**, then stages a fully processed copy for downstream list-building scripts.

Key Responsibilities

- Sheet creation and reuse (“AR filters”, “Mike’s list”, “Team’s list”)
- Structural cleanup (row and column pruning)
- Logic-based row deletion
- Enrichment via lookup formulas
- Visual formatting for analyst usability
- Creation of a reusable AR decision flag

Exceptionally Clever Logic

1. Column-Name Driven Deletions

Instead of hard-coding column positions, the script dynamically locates columns like "FC", "State", and "Stu Num" by header name.
This makes the automation resilient to schema drift in upstream exports.

2. Bottom-Up Row Deletion Strategy

Rows earmarked for deletion are collected first, then deleted **from bottom to top**, preventing index corruption—an easy mistake in Excel automation that often leads to missed or unintended deletions.

3. Selective VLOOKUP Injection

The script only inserts lookup formulas into the “Notes” column **when a key field is populated**, preventing unnecessary formula noise and improving performance on large datasets.

4. Value-Freezing Before Sorting

Dates are reformatted, copied as values, and only then sorted.
This avoids Excel recalculation issues that can silently corrupt sort order when formulas remain live.

5. Derived AR Decision Column

A helper column (“AR ≥ 100”) is inserted programmatically and populated using relative R1C1 logic.
This transforms numeric balances into a **binary decision flag**, dramatically simplifying downstream filtering logic.

Script 2 — Personal Outreach List Builder (AR + Non-AR)

(Mike's List V1–V3 logic)

Purpose

This script builds a **single, email-ready outreach list for Michael Larrauri**, combining multiple eligibility rules and message types into one controlled output.

Key Responsibilities

- Pull from the staged "AR filters" dataset
- Build AR and non-AR segments sequentially
- Append, not overwrite, outreach lists
- Apply visual segmentation and messaging logic
- Update source dashboard contact dates

Exceptionally Clever Logic

1. Logic-Driven Filtering (Not Visibility-Driven)

Although Excel filters are applied for analyst visibility, **all record selection is performed in code**, not by reading visible rows.
This prevents errors caused by filtered-out rows still existing in memory.

2. Append-Only Architecture

Non-AR outreach records are appended below AR records rather than written to a separate list.
This produces a **single, ordered, send-ready file** while preserving segment identity.

3. Color-Coded Semantic Meaning

- Yellow rows = AR outreach
- Green rows = non-AR outreach

Visual meaning is encoded directly into the dataset, improving usability without requiring documentation.

4. Template Decoupling

Email subject lines and HTML bodies are read from static template cells.
Updating messaging requires **no code changes**, only editing a worksheet cell.

5. Bidirectional Automation

The script doesn't stop at list creation—it writes the current contact date **back into the source dashboard**, closing the loop between analytics and operations.

Script 3 — Team Outreach List Builder with Volume Control

Purpose

This script builds a **team-wide outreach list** with strict prioritization and volume enforcement, ensuring fairness, freshness, and compliance with daily sending caps.

Key Responsibilities

- Exclude Michael Larrauri's records
- Prioritize uncontacted ("new") records
- Backfill older records if needed
- Enforce a hard cap (e.g., 120 emails)
- Populate email templates automatically
- Update contact dates in the dashboard

Exceptionally Clever Logic

1. Priority Tiering

The script intentionally splits records into:

1. New/uncontacted records
2. Older contacted records

New records are always selected first, ensuring **maximum operational impact per send**.

2. Backward Iteration for Backfill

Older records are collected from the **bottom of the dataset**, ensuring the oldest contacts are refreshed first instead of cycling recent entries repeatedly.

3. Single-Source Cap Enforcement

The email cap is calculated once and enforced across both tiers, guaranteeing the final list never exceeds the target while still maximizing priority coverage.

4. Two-Phase Dashboard Update

Contact dates are updated in the same priority order as list construction, ensuring the dashboard reflects **exactly what was sent**, not just what was eligible.

Script 4 — VBA Automation (Legacy/System Bridge)

Purpose

The VBA component serves as a **bridge automation layer**, enabling compatibility with environments or user actions not yet supported by Office Scripts alone.

Role in the Architecture

- Completes end-user execution flows
- Triggers or complements Office Script logic
- Ensures the solution remains viable in mixed Excel environments

This hybrid approach reflects real-world production constraints and demonstrates architectural pragmatism rather than theoretical purity.